Raffaello Fornasiere

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Summary

I am an AI safety researcher specializing in LLM interpretability and training methodologies. I am currently at **LASR Labs**, stress-testing whether current LLM interpretability methods remain reliable on models trained with different methods. I have a background in computer science and have previously worked as a lead software engineer and AI integration developer.

Professional Experience

AI Safety Researcher, LASR Labs (April 2026 - August 2026)

- My team at LASR Labs has been recently awarded a grant of £200k from Coefficient Giving to extend the project for 4 additional months, continuing under the mentorship of **Stefan Heimersheim**.
- Investigating which training variations of model organisms cause mechanistic interpretability methods to fail, to expose blind spots in current auditing benchmarks. Our **paper** has accepted at ICML Interpretability Workshop 2026.

Technical AI Safety course facilitator, BlueDot Impact (May 2026 - June 2026)

- Part-time (7h/week) facilitating two cohorts through BlueDot Impact's Technical AI Safety course, leading weekly discussions on AI Safety.

LASR Labs Research Fellow, LASR Labs (January 2026 - April 2026)

- Selected for a 13-week AI safety research program in London.
- Working in a team mentored by **Stefan Heimersheim** to stress test LLM interpretability methods under different training methodologies.

ARENA 7.0 Mentee, ARENA (January 2026 - February 2026)

- 5-week full-time bootcamp focused on ML engineering for technical AI safety.
- Working on transformers architecture, reinforcement learning, mechanistic interpretability, and safety evaluations.
- Awarded 1st prize at ARENA's interpretability hackathon.

AI Agent Developer, Freelance for William Saunders, Alignment Science Researcher at Anthropic (May 2025 - December 2025)

- Built a multi-agent AI assistant that maintains coherent context over months of use, avoiding the repetitive patterns typical of commercial assistants.
- Integrated tool add-ons including asynchronous browser use, Google Calendar, and speech-to-speech interfaces.
- Tech stack: FastAPI, React, Claude API, RAG, event-driven architecture.

Lead Software Engineer - Front-End, Infiniteloop (October 2022 - August 2025)

- Prototyped AI healthcare applications using LLMs, exploring feasibility for medical report automation (related to my thesis work).
- Designed, built and maintained user interfaces using Angular, ensuring seamless integration with the JHipster framework.
- Developed reusable JHipster framework extensions (e.g., an extended JHipster Criteria API, a custom Spring Data JPA repository for bidirectional relationship synchronization).
- Led the development of multiple products.
- Mentored two junior developers and promoted best practices to improve code quality.

Junior Full Stack Developer, Prodigys Group (July 2021 - September 2022)

- Developed and maintained both front-end and back-end components using Angular, SpringBoot, and Django.
- Optimized database performance through SQL query refactors, reducing execution times from hours to minutes.

Education

Master's in Computer Science and Engineering, Politecnico di Milano (Graduated July 2024)

- Thesis: Exploring the Potential of Lightweight LLMs for Medication and Timeline Extraction.
- Developed a web application integrating LLMs for medical information extraction, **tested in three Italian hospitals**.
- Published research: "Medical Information Extraction with Large Language Models" (ICNLSP-2024, ACL Anthology).

Bachelor's in Electronics Engineering, Università degli Studi di Udine (Graduated July 2020)

- Thesis: Optimization of Digital Circuit Propagation Times Using Genetic Algorithms.

Technical Skills

ML Research & Engineering:

- Model training and fine-tuning (SFT, DPO) up to 1B parameters, data pipeline construction and filtering at scale.
- Linear probes and classifiers for mechanistic interpretability, LLM-as-judge labelling pipelines.
- Transformer internals (TransformerLens, activation hooks), RL and RLHF, safety evaluations and benchmarking.
- Libraries: PyTorch, Hugging Face Transformers, TransformerLens, llama.cpp.

AI/LLM Integration:

- LLM APIs (OpenAI, Anthropic Claude), agent development, RAG, prompt engineering, local models (llama.cpp).

Languages:

- Primary: TypeScript/JavaScript, Python, Java.
- Also experienced with: C++, SQL.
- Others: CSS, HTML.

Frameworks, Libraries and Tools:

- Back-end: FastAPI, Django, Spring Boot, JHipster.
- Front-end: React, Vue.js, Angular.
- Others: Docker, Git, Tailwind CSS, Streamlit, Spring Data JPA, Postgres/PostgreSQL, SQLite.

Publications

The Model Organism Lottery: Model Organism Interpretability Strongly Depends on Training Methodology
Published at ICML Interpretability Workshop 2026
`https://arxiv.org/abs/2607.01033`

- Authors: Andrzej Szablewski, Gabriel Konar-Steenberg, Raffaello Fornasiere, Nikita Menon, Stefan Heimersheim

Medical Information Extraction with Large Language Models
Published at ICNLSP-2024, ACL Anthology
`https://aclanthology.org/2024.icnlsp-1.47/`

- Authors: **Raffaello Fornasiere**, Nicolò Brunello, Vincenzo Scotti, Mark Carman.

Academic Projects

Medical Image-Caption Matching with Transformers

- Reimplemented CLIP model from scratch for medical radiology applications using ResNet vision encoder and custom transformer text encoder.
- Trained and fine-tuned both models using contrastive learning, pre-trained BERT integration, caption preprocessing, and medical concept augmentation techniques.
- Developed evaluation methods for assessing image-caption alignment in medical domain.

Online Learning for Dynamic Pricing in E-commerce

- Exploration work implementing multi-armed bandit algorithms (GP-TS, GP-UCB) with social influence modeling to optimize advertising budget allocation across product campaigns.

Deep Learning Projects with CNNs

- **Species Classification:** Implemented transfer learning with VGG16/19 and EfficientNet models, applying data augmentation techniques (CutMix, MixUp) and ensemble methods to achieve 86.91% accuracy on image classification tasks.
- **Time Series Classification:** Developed CNN-LSTM hybrid architectures for sequential data analysis, experimenting with preprocessing techniques including normalization and noise augmentation to reach 70% classification accuracy.

Other Experiences

STAIR Mentor, St Andrews Impact Research (Feb 2026 - March 2026)

- Mentoring 2 early-career students beginning AI safety research.

ML4Good AI Safety Bootcamp

- Completed EPFL's iteration of the ML4Good AI Safety and ML Bootcamp (February 2023).

Community Involvement

Volunteer developer for hometown community

- Built website to publish local competition status with real-time updates through Telegram bot.
- Developed web application to emulate Italian TV game show, managing multiplayer gameplay across mobile devices.